Haozhe Liu

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Efficient AI, NVIDIA Research

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Research Scientist at NVIDIA Research, working on efficient and scalable AI, with hands-on R&D experience in large-scale multimodal generation across image and video. Previously worked with Meta GenAI / BizAI, contributing to the successful delivery of ImageGen and VideoGen products and large-scale foundation-model systems. Experienced in 1K+ GPU-scale pretraining, text-to-image generation, image-to-video generation, autoregressive diffusion, and efficient generative modeling. First-authored 10+ top-tier publications with 1,000+ citations, and received a Workshop Best Paper Award. Received a Ph.D. in Computer Science from KAUST, advised by Prof. Juergen Schmidhuber.

EDUCATION

- **KAUST – AI Initiative** — Saudi Arabia
  Ph.D. in CS (Advisor: Juergen Schmidhuber) — *Aug. 2022 – Mar. 2026*
  - GPA: 4.0/4.0
  - Research Interests: Explainable, Efficient, and Responsible Generative Models

SELECTED PUBLICATIONS

1. **Liu, H.**, Liu, D., Zhuge, M., Zhou, Z., Xie, T., He, S., et al. & Schmidhuber, J. Mixture of States: Routing Token-Level Dynamics for Multimodal Generation. CVPR 2026. [pdf]
2. **Liu, H.**, Liu, S., Zhou, Z., et al. & Perez-Rua, J. M. MarDini: Masked Autoregressive Diffusion for Video Generation at Scale. TMLR. [pdf]
3. **Liu, H.**, Zhang, W., Xie, J., Faccio, F., Xu, M., Xiang, T., Shou, M., Perez-Rua, J. M. & Schmidhuber, J. Faster Diffusion Through Temporal Attention Decomposition. ICLR 2026 & TMLR. [pdf] [code]
4. **Liu, H.**, Zhang, W., Li, B., Ghanem, B. & Schmidhuber, J. "Lazy" Layers Make Fine-tuned Diffusion Models More Traceable. Technical Report. [pdf]
5. **Liu, H.**, Zhuge, M., Li, B., Wang, Y., Faccio, F., Ghanem, B. & Schmidhuber, J. Learning to Identify Critical States for Reinforcement Learning from Videos. ICCV 2023. [pdf] [code]
6. **Liu, H.**, Zhang, W., Li, B., Wu, H., He, N., Huang, Y., Li, Y., Ghanem, B. & Zheng, Y. AdaptiveMix: Improving GAN Training via Feature Space Shrinkage. CVPR 2023. [pdf] [code]
7. **Liu, H.**, Li, B., Wu, H., Liang, H., Huang, Y., Li, Y., et al. & Zheng, Y. Combating Mode Collapse in GANs via Manifold Entropy Estimation. AAAI 2023. [pdf] [code]
8. **Liu, H.**, Wu, H., Xie, W., Liu, F. & Shen, L. Group-wise Inhibition-based Feature Regularization for Robust Classification. ICCV 2021. [pdf] [code]
9. **Liu, H.**, Zhang, W., Liu, F., Wu, H. & Shen, L. Fingerprint Presentation Attack Detector Using Global-Local Model. IEEE Transactions on Cybernetics. [pdf] [code]
10. **Liu, H.**, Zhang, W., Xie, J., Wu, H., Li, B., Zhang, Z., Li, Y., Huang, Y., Ghanem, B. & Zheng, Y. Decoupled Mixup for Out-of-Distribution Visual Recognition. ECCV 2022 Workshop. [pdf] [code]
11. Zhang, W., **Liu, H.**, Li, B., et al., Zheng, Y. & Ghanem, B. Dynamically Masked Discriminator for Generative Adversarial Networks. NeurIPS 2023. **Equal Contribution**. [pdf] [code]
12. Zhang, W., **Liu, H.**, Liu, F., Ramachandra, R. & Busch, C. Effective Presentation Attack Detection Driven by Face Related Task. ECCV 2022. **Equal Contribution**. [pdf] [code]
13. Zhuge, M., **Liu, H.**, et al. & Schmidhuber, J. Mindstorms in Natural Language-Based Societies of Mind. NeurIPS Workshop 2023. **Best Workshop Paper, Equal Contribution**. [pdf] [code]

AWARDS, GRANTS & HONORS

Workshop Best Paper Award 2023
Co-PI, "Efficient Generative Models", SDAIA–KAUST Grant (**200k SAR**) 2023
Outstanding Graduate Award, SZU (**Rate** $\leq$ **5%**) 2022
China National Scholarship, SZU (**Rate** $\leq$ **0.02%**) 2021

RESEARCH EXPERIENCE

- **Efficient AI, NVIDIA Research** Singapore
  Research Scientist working with Prof. Song Han *2026 – present*
  - Research Topic: Efficient AI, large-scale pretraining, image/video generation.
- **BizAI, Meta** MPK, United States
  Research Scientist Intern working with Ding Liu and Wei Liu *2025*
  - Research Topic: Large-scale pretraining, text-to-image generation, mixed-modality generation.
  - Publication Records: CVPR $\times$ 4.
  - Co-developed a commercial 7B text-to-image model trained on over 1B image–text pairs and deployed in several widely used products.
  - Built a 20B native LLM-driven multimodal generation prototype from scratch.
  - Contributed to a native unified model for both visual understanding and generation tasks.
  - Contributed to Neural Computer Project.
- **MGenAI, Meta** London, United Kingdom
  Research Scientist Intern working with J.-M. Perez-Rua and T. Xiang *2024*
  - Research Topic: Foundational training, image-to-video generation, text-to-image generation.
  - Publication Records: TMLR $\times$ 2; ICCV $\times$ 1; CVPR $\times$ 1.
  - Co-developed a foundational text-to-image model to support several well-known products.
  - Scaled autoregressive diffusion to video generation.
- **GenAI, KAUST** Saudi Arabia
  Ph.D. Candidate supervised by Prof. Juergen Schmidhuber *2022 – 2026*
  - Research Topic: Neural networks with multiple-step inference, including diffusion models, autoregressive models, and RL agents.
  - Publication Records: ICCV $\times$ 1; CVPR $\times$ 2; NeurIPS Workshop $\times$ 1; TMLR $\times$ 2; under review $\times$ 1.
  - Highlight: NLSOM received a **Workshop Best Paper Award**.
  - Highlight: T-GATE was merged into the Diffusers library and received over **300** stars on GitHub.
- **Jarvis Lab, Tencent** Shenzhen, China
  Intern working with N. He, Y. Li, and Y. Zheng *2021 – 2022*
  - Research Topic: Generative models and medical imaging.
  - Publication Records: NeurIPS $\times$ 1; CVPR $\times$ 1; ICCV $\times$ 1; AAAI $\times$ 1; MICCAI $\times$ 2; MIA $\times$ 1; PR $\times$ 1; ECCVW $\times$ 1.
  - Highlight: MaF-GAN was recognized as an **Oral** paper at AAAI.
  - Highlight: Ranked 4th in the ECCV 2022 NICO Challenge.